

Lesson 24: The Wave Equation - Multidimensional Waves & Dispersion

Professor Dave Explains Differential Equations Master Series · Lesson 24

Video: The Wave Equation Part 2: Multidimensional Waves and Dispersion Course: Applied Ordinary Differential Equations (ENGR 213)

1. Video Intuition & Pedagogical Framing

Professor Dave expands the wave equation into two and three dimensions: $u_{tt} = c^2 \nabla^2 u$. When vibrating a circular drumhead, Cartesian coordinates fail, and cylindrical polar coordinates summon Bessel functions $J_m(kr)$, producing beautiful circular and radial nodal lines where the drumhead remains motionless. Furthermore, Dave explains wave dispersion: in real physical media (like ocean waves or optical fiber glass), different frequencies travel at different speeds, causing phase velocity $v_p = \omega/k$ to detach from group velocity $v_g = d\omega/dk$!

2. Core Theoretical Framework & Rigorous Formulations

1. The 2D Wave Equation in Polar Coordinates

$$\frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} \right)$$

For axisymmetric vibrations ($u = u(r, t)$): spatial separation yields **Bessel's Equation of Order Zero**:

$$r^2 R'' + r R' + k^2 r^2 R = 0 \implies R(r) = J_0(kr)$$

Clamped edge condition $R(a) = 0 \implies k_{0,n} = \frac{\alpha_{0,n}}{a}$, where $\alpha_{0,n}$ are the zeros of $J_0(z)$.

2. Dispersion Relations: Phase vs Group Velocity

$$\omega = \omega(k) \quad (\text{Dispersion Relation})$$

- Phase Velocity:** $v_p = \frac{\omega}{k}$ (speed of individual wave crests).
- Group Velocity:** $v_g = \frac{d\omega}{dk}$ (speed of the overall wave packet and energy transfer).
- If $v_p \neq v_g$, the medium is **dispersive**, causing pulses to broaden and distort over time.

3. Fully Solved Whiteboard Problems

Optics Problem: Calculating Phase vs Group Velocity in a Dispersive Medium

Problem Statement: In a plasma waveguide, electromagnetic waves obey the dispersion relation $\omega(k) = \sqrt{c^2 k^2 + \omega_p^2}$, where ω_p is the plasma frequency. Compute the phase velocity v_p and group velocity v_g , and show that $v_p \cdot v_g = c^2$.

Step 1: Compute Phase Velocity

$$v_p = \frac{\omega}{k} = \frac{\sqrt{c^2 k^2 + \omega_p^2}}{k} = \sqrt{c^2 + \frac{\omega_p^2}{k^2}}. \text{ Notice } v_p > c \text{ (individual wave crests travel faster than the speed of light in vacuum!).}$$

Step 2: Differentiate Dispersion Relation to Compute Group Velocity

$$\omega^2 = c^2 k^2 + \omega_p^2 \implies 2\omega \frac{d\omega}{dk} = 2c^2 k \implies \frac{d\omega}{dk} = \frac{c^2 k}{\omega}.$$

Step 3: Evaluate Group Velocity v_g

$$v_g = \frac{c^2 k}{\sqrt{c^2 k^2 + \omega_p^2}} = \frac{c^2}{v_p}. \text{ Notice } v_g < c \text{ (physical energy and information travel strictly subluminally, preserving Einstein's special relativity!).}$$

Step 4: Verify Product Relation

$$v_p \cdot v_g = v_p \left(\frac{c^2}{v_p} \right) = c^2.$$

⚠ Common Exam Pitfall & Trapdoor Avoidance

Never claim that physical signals travel faster than light when $v_p > c$. Information and energy travel at the group velocity v_g , which strictly obeys $v_g \leq c$.

4. Key Takeaways & Exam Strategy Formula Card

 Core Takeaway & Exam Rule

Circular membranes vibrate in Bessel modes $J_m(kr)$. Dispersion ($d\omega/dk \neq \omega/k$) separates individual phase crests from overall packet energy.

- Video reference: [The Wave Equation Part 2: Multidimensional Waves and Dispersion](#)
- Always verify boundary conditions and check that candidate solutions satisfy the governing differential equation identically.